

Visions of Sound and Colour: Generating Visuals and Sound from a Source Photographic Image

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Abstract

Presenting developments of previously published work,¹ this chapter is concerned with the expression of creative subjectivity through computer code. It argues that today, computer code is a medium available for art practice and visual research, as able as any other more conventional medium to convey artistic subjectivity. A brief introduction is given to the author's main research project — personal visual languages for artistic expression — within which a working definition of 'personal visual language' is proposed in order to clarify the terms and concepts discussed within the argument. Relevant features of the author's personal visual language are identified, and the translation of these features into the code itself is discussed. It is argued that computer code, as written by the artist, is an intrinsic part of the final artwork and not just an application used by the artist to generate his/her artwork. The author's ongoing visual project is given as an example to explain the concepts discussed within the chapter. Following the constraints of a series of rules imposed within the code, the discussed algorithm captures the light, colour and general atmosphere of a source photographic image and, through a process of decision making, progressively translates these features into a time-based piece of work running real time. The algorithm generates a structured composition, at the same time geometrically organized and organically chaotic, following the pixel's colours of the source image. It also generates sounds based on the HSB values of these colours, playing a 'melody', which is unique for each generated composition. A general description of the algorithm developed for the generation of the sound is given. It is expected that any conclusions driven from this research will lead to a better understanding of the creative process within the context of code based visual work.

Key Words: Code art, software art, algorithmic art, fine arts, digital arts, digital music, algorithmic music, generative art, generative music, digital age, visual expression, visual language, artistic expression, landscape art, soundscapes.

1. Introduction

The methods and strategies followed by the artist to confer individuality and expressive subjectivity to his/her visual work can be referred to as personal visual languages, a concept derived from the generally accepted view of visual languages as systems of visual signs. In the context of this chapter, and in order to clarify this concept, the following definition is proposed: a personal visual language is the

collection of visual features, aesthetic solutions and visual strategies that characterizes the visual work of an artist.

A personal visual language manifests itself within the visual work of an artist as 'the style of the artist', as E. H. Gombrich puts it, resulting in a subjective transformation of the subject matter. Although Gombrich never refers to it as a personal visual language, he is very clear in his analysis of this phenomenon, stating that the artist's 'selective preferences' are at the origin of such transformation.²

The features of a personal visual language can be seen as personalized versions of the features of artistic languages in general. Artist-dependent components such as the artist's elected materials and techniques, and his/her choice of palette and subject matter, for instance, can be considered. A personal combination of features can lead to the specific aesthetic solutions and visual strategies that characterize the work of an artist.

Since the 1980's, graphic software and graphic interfaces that mimic conventional tools and techniques have been developed. These graphic applications are proposing a range of new methods and techniques that extend and enhance those of conventional art and design practices. Some of these tools and methods are designed to be so accurately similar to their conventional counterparts that they don't really represent a paradigm shift within the field of art making. The same cannot be said when the artist's visual practice is conducted by writing code. Code based art practices represent an undeniable paradigm shift, demanding a new approach to visual thinking and visual research. The algorithm that is discussed in this chapter was developed as an artwork, conveying an aesthetic vision. It is not a general-purpose application, and is not conceived to function as such. The aim of this chapter is to discuss the idea that the algorithm is translating the author's personal visual language as efficiently as any other media would.

A brief outline of the author's visual project and personal visual language is given first, as an introduction to the discussion that follows.

2. Visual Project, Visual Language

As a result of a lifelong attraction to natural environments, the author's personal visual project aims at translating into visual form the subjective impression of clarity, purity and beauty she perceives in a landscape under specific atmospheric conditions of light and colour. It deals also with the role played by randomness on the construction and deconstruction of complexity and diversity on natural environments and on the artwork they inspire.

The project went through different mediums, materials and techniques since the 70's, when the first attempts were made at capturing random variations of light and colour in the landscape through photography. The computer came into the project as a tool and a medium in 1984, and randomness could be visually explored in full as an aesthetic component. Photographic images remained to this day the main source of material for all the artwork developed within the ambit of this project.



Figure 1: *Pico Grande*, 1985, ZX Spectrum 48K (black & white prints of still frames) © 2015. Courtesy of Evangelina Sirgado de Sousa.

The features of the author's personal visual language can be briefly outlined as follows: nature, natural environments and natural phenomena as subject matter; non-realistic combinations of otherwise realistic elements; when time-based, running in an endless loop; the pace is set calm and slow, to invite and induce a contemplative attitude towards the subject matter; the poetics of nature's colours and constantly changing compositional situations is visually explored.

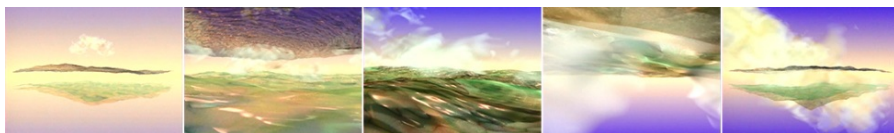


Figure 2: *Three Exercises on Landscapes (and beyond)*, 1993, frames of 3D animation.³ © 2015. Courtesy of Evangelina Sirgado de Sousa.

In different combinations, these defining features are manifest throughout all the body of work, independently of the specific medium, technique and/or materials in which the artwork took form at a given time. (Fig. 1, 2 and 3)

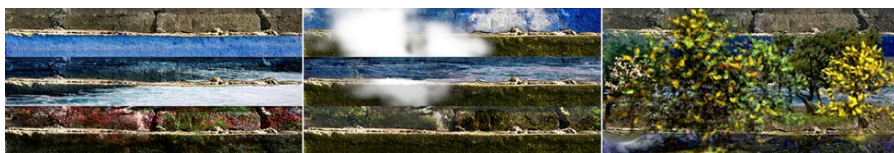


Figure 3: *The Opposite Side*, 2010, screen capture of Flash interactive animation. © 2015. Courtesy of Evangelina Sirgado de Sousa.

3. Encoding Ideas

When collecting photographic material for a project, colour is an element of paramount importance as a major contributor for the desired aesthetic result. Ultimately a photograph is just an array of colour pixels. At what extent an abstract image composed exclusively around the colours of a given photograph would convey the actual feelings/impressions/motivations that conditioned the selection of the image?

Encoded Thoughts is the ongoing stage of the author's visual project. It runs in two versions: an interactive version that has been discussed briefly in a previous paper,⁴ and this version, designed to run endlessly on a computer and encourage contemplation. Both versions are conceived to generate countless abstract versions of a given photographic image, translating the author's subjective impression of the source image exclusively through its colour and luminosity, (almost) completely ignoring its representational nature.

Computer code was the creative medium elected to translate these ideas. It can capture accurately the colour information of individual pixels and generate both, still and moving images. But can it express subjective creativity? Between the artist and his work there is always a medium through which the artwork materializes itself. The artist has to master his/her elected medium and technique before he/she can 'speak' through it. Computer code can be seen as an immaterial medium, one that requires logical and systematic thinking. When coding, there is no direct manipulation of materials, no artistic hand skills involved, and therefore, (apparently) no room for creative spontaneity. Can a personal style or a personal visual language develop in such circumstances? The proposed answer is 'yes', and this is the reason: in order to develop a personal style, human factors have to be involved. There is an obvious human factor involved in programming—the actual programmer (a human being) behind the code. If he/she is also an artist, and if the code is developed specifically as an art practice, the usual creative process will naturally take place.

Let's try and define briefly what is to be encoded in this case: the induction of peaceful minds and the expression of a personal, subjective perception of beauty in nature. The algorithm will accomplish this by generating compositions with a strong vertical and horizontal structure — vertical/horizontal patches or a combination of the two, fading to the borders: peace, serenity, stability, but also spiritual elevation. Randomness is strongly at play leading to a controlled unpredictability (controlled chaos in nature). Colour and transparency contributes to a gentle, soft, soothing effect. Colours are captured from a source photographic image, carefully selected having in mind the desired effect. The picked colours appear within the composition at their original relative positions, centred at the pixel where the colour was picked from.



Figure 4: Three generated images and its source photographic image.
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The size and transparency of the patches, as well as the rhythm of depiction, are controlled by a process of decision making based on the degree of brightness/darkness of the colour (darker colours generate smaller, more transparent patches); the delay between darker patches is longer; a succession of brighter patches will be displayed at a faster pace, and the patches will be larger and less transparent. The overlapping of semi-transparent patches contributes to the generation of the desired soothing, calming effect. For a couple of minutes, the algorithm will follow these procedures until a ‘final’ image is saved as a still frame. The algorithm will then select another source image and repeat the process, *ad aeternum*. The result is an animated piece of work, running real-time and continuously generating new aesthetic situations, inducing contemplation and never repeating itself. The running process is actually more important than the final, saved image –which, many times, is not at all as interesting as the aesthetic situations generated along the way.

The actual code writing is a dynamic process of trial and error and continuous experimentation, far from the systematic approach usually taken when writing a conventional application. Artist’s code is never ‘ready’ — it develops following the author’s experimental path, leading to spontaneous and rather unexpected results that can then be purposely included within the piece. The following image shows a result of an alternative version of the algorithm, which draws partial samples of the source image into the otherwise abstract image.



Figure 5: Image generated by a second version of the same algorithm.
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4. Sound

Encoded Thoughts was initially conceived as a purely visual piece having no sound. When the need for adding sound came up, and since all versions of this piece are generative and run in real time, it was clear that the sound had to be generative as well, and a very close connection between sound and image had to be kept. Two different concepts were considered and implemented:

a) Based on the RGB Colour Model,⁵ the sound is generated by mapping independently the red, green and blue values of the pixel's colours to musical notes spanning over (approximately) five octaves, based on the brightness of the colour (lower pitches for lower brightness values). The three notes are then played simultaneously as a chord.⁶

b) Based on the HSB Colour Model,⁷ the sound is generated by mapping the actual hue of the colour to specific notes based on Newton's colour wheel, and the saturation and brightness values into musical notes spanning over (approximately) five octaves. The three notes are then played simultaneously as a chord.⁸

A brief description of the methods and their implementation are given here.

4.1 RGB

As it is well known, a colour (seen as light) can be described in terms of its three primary components, red, green and blue. A mix of the three components at its lowest intensity will produce black, and at its full intensity will produce white.

Specific percentages of each component, additively mixed, produce a specific colour. The intensity — or the amount — of each component is measured in a scale from 0 to 255 (256 values). The first implementation of this algorithm attempted at mapping each one of these values to a musical note, being the lower values mapped onto the lower pitch notes. It became clear at once that the lower pitch notes would be far too low and the same was true for the higher pitch notes that would come up rather squeaky. The number of mapped values had to be reduced to a manageable but consistent amount. Since many modern pianos have 88 keys (seven octaves plus a minor third, from A0 to C8), it makes sense to keep the number of mapped notes well within this range, but this number had also to be consistent with the 256 values they are supposed to be mapped into. After considering different possibilities, this was achieved the simplest way: by dividing 256 by 4, reaching a number of 64 notes to be mapped into 256 values. This would allow each note to be mapped into four R, G or B consecutive values.

Once this decision was made, the next step would be to define the range of musical notes into which to perform the mapping. It was decided by trial and error to start at C1 (MIDI 24)⁹ for the lowest four values and going up until E6 (MIDI 88) (5 octaves + 4 notes). Values of 0, 1, 2 and 3 are mapped into MIDI 24; values of 252, 253, 254 and 255 are mapped into MIDI 88. As an example: RGB (223, 102, 93) will play G#5, D3, B2 (MIDI note numbers 80, 50, 47).¹⁰

Although this implementation produced interesting and pleasant results, it wasn't entirely satisfactory. There isn't a real relationship between the actual colour and the sounds produced. The relationship is towards the independent Red, Green and Blue values, not the three together as a colour. For instance, the colour RGB (223, 102, 93) — a very nice pink — will play exactly the same chord as the colour RGB (102, 223, 93)—a very nice green.

4.2 HSB

The second implementation takes into account the actual hue of the colour, mapping the 12 notes of an octave into the 12 primary, secondary and tertiary colours of a colour chart.

The HSB Colour Model is just a different way of representing the RGB colour model, taking into consideration not only the actual hue of the colour (often arranged in a circle and numbered from 0 to 360) but also the brightness and saturation of the colour (measured in a scale from 0 to 100). (Fig 6) Again, the mapping of 360 musical notes into the 360 hues of the colour model is not feasible, since it would generate unplayable low and high pitches.

It is interesting to realize that the colour chart has the same number of notes as an octave. This fact lead to an obvious solution: to attribute each one of the 12 notes of an octave to a colour in the chart. But this 'simple' solution proved to be more complex than expected. Which colour to which note?

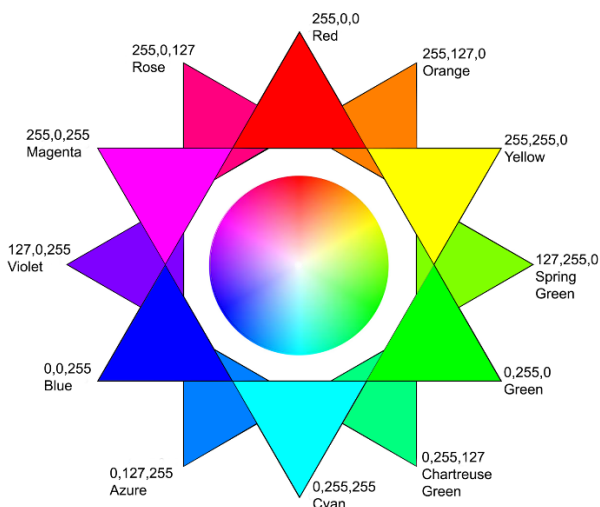


Figure 6: Colour chart, showing the primary, secondary and tertiary colours, its respective name and RGB values. The circular representation in the middle shows the 360 hues of the HSB model. © 2015. Courtesy of Evangelina Sirgado de Sousa.

Although a final conclusion has never been reached, the correlation between colour and sound has been vastly explored and researched since Newton's famous experiments with the decomposition of white light. A simple Internet search will lead to countless sites on the subject, presenting solutions based on mystical, spiritual, and vaguely scientific reasoning—and in complete disagreement with each other. A more consistent line of inquiring would be the search for a correspondence between the frequency or the wavelength of colour and sound—an inquiry that clashes on the simple fact that the frequency of light is significantly higher than the frequency of sound. Nevertheless, this line of inquiry will be a next step for this research project. For the time being, it was decided to experiment with Newton's primordial attempt at correlating colour and sound.

Based on his knowledge of music, Newton attempted a correspondence between the colours of his recently discovered spectrum (red, orange, yellow, green, blue, indigo and violet) and the seven notes of an octave (ignoring sharps and flats).¹¹ According to Jeffrey N. Howard, 'thus was born the assumed relationship of "do, re mi, fa, sol, la, ti" with the seven visible spectrum colours ROYGBIV—a relationship that even Newton himself never fully explained'.¹² (Fig. 7)

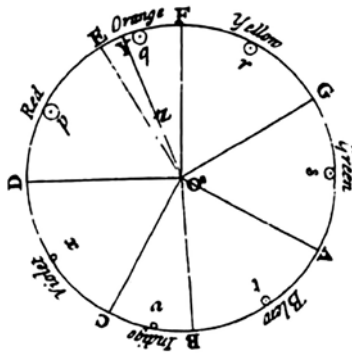


Figure 7: Newton's colour wheel (from Newton, 1704, book I, Part II, Plate III).

© Image obtained from Project Gutenberg and is in Public Domain.¹³

As many have done before, let's superimpose Newton's wheel over the HSB colour chart, aligning the hues as close as possible. If we align the yellow sector of Newton's wheel with the correspondent chart sector, we can see that practically all other colours fall into place. Although the alignment of the red sector is somewhat loose, the orange, green, blue, indigo and violet sectors are aligned (indigo is an ultramarine blue). This places perfectly the seven notes indicated in Newton's wheel over the azure, blue, magenta, red, orange, spring green and chartreuse green (C is Middle C). And, in this experiment, the remaining colours became the correspondent sharp notes. (Fig 8)

In order to consider the complete circle of 360 hues, to each note is assigned 30 hues (360/12). Considering the position of the red colour as 0, E will be assigned to positions 246 to 15 within the colour wheel (30 shades of red), F will be assigned to positions 16 to 45 (30 shades of orange), and so on. To cause the notes to span over 5 octaves, the brightness of the colour is taken into consideration. For instance, if the brightness of the colour is a value between 0 and 20 (100/5), the MIDI number of the note is subtracted by 36 (3 octaves down); if the brightness is a value between 80 and 100, 12 is added to the note's MIDI number (one octave up). The note assigned to the colour's hue is played as a 'solo' (Pan Flute for higher pitches, Music Box for very low pitches with which the Pan Flute cannot deal).

To the saturation and brightness values of the colour, the same mapping system already described for the RGB algorithm was applied. Considering brightness and saturation values between 0 and 100, values of 0 are mapped into MIDI 24, values of 100 are mapped into MIDI 88. The two notes for saturation and brightness and the hue's 'solo' are then played simultaneously as a chord, coinciding with the generation of the correspondent patch of colour within the composition.

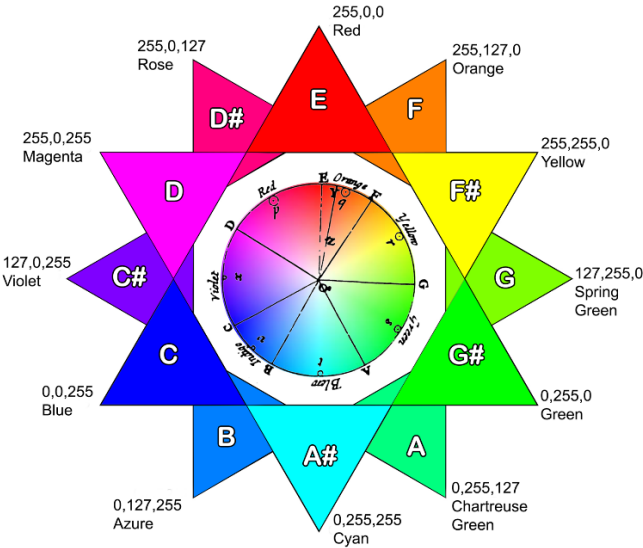


Figure 8: Colour chart, showing the superposition of Newton’s colour wheel over the circular representation of the 360 hues of the HSB model.
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This method leads definitely to a better relationship between the sound and the actual colours. The assignment method is more precise and accurate, and leads to a better translation of the general mood of the source image.

To generate musical sentences and avoid a mechanical play, some rhythm was introduced within the code, as follows: since the patches are deployed in groups of a random number of elements, a delay was set between patches, and between groups of patches. The values of these intervals are kept within specific limits, and are determined by the brightness of the patch colour. Higher brightness values will cause the patches to be displayed with shorter delays in-between. The delay between groups of patches is a longer Gaussian random value, separating visual and musical sentences. Also, brighter colours play louder and very dark colours are barely audible, with the intensity (or loudness) of sound mapped into the brightness value of the colour. This ‘tempo’ method was implemented on both, RGB and HSB models.

5. Conclusions and Further Work

Coding as a form of artistic expression is a completely different matter than coding with more technical purposes in mind. The resultant code is not intended as

an application for general use: it is an integral part of the final art piece. The approach to coding as an artist is free and loose and open to spontaneous improvisation along the process, starting with a loosely defined idea and developing according to an artistic goal. The resultant piece is not entirely predictable and can take many turns along the way. The algorithm briefly described in this chapter is one of the many possibilities opened up by the initial proposed ideas. It could have evolved differently. Other artist would interpret the initial goals in a different way, according to his/her own aesthetic vision and ‘speaking’ their own personal visual language.

Although *Encoded Thoughts* is now a fully developed piece, there is still some room for improvement. There is, specifically, two points that need further research. At what extent can the sound/colour relationship be approached from other points of view? Is it possible to transcend the 12 notes of an octave and generate a larger range of sounds, in order to cover a larger number of colours within the HSB colour chart? The possibilities are endless. Eventually, this visual project may end up evolving in a completely different and unpredictable direction...

Notes

¹ Evangelina Sirgado de Sousa, ‘Encoded Thoughts: Writing Code as an Art Practice’, *Arts and Technology: Third International Conference, ArtsIT 2013*, ed. Giorgio De Michelis et al. (London: Springer, 2013), 89, viewed 26 October 2015, http://link.springer.com/chapter/10.1007%2F978-3-642-37982-6_12#page-1.

² E. H. Gombrich, ‘Truth and the Stereotype’, *Art and Illusion* (London: Phaidon Press, 2002), 55.

³ Evangelina Sirgado de Sousa, *3 Exercises on Landscapes and Beyond* (video), viewed 26 October 2015, <https://vimeo.com/90958189>.

⁴ Sirgado de Sousa, ‘Encoded Thoughts: Writing Code as an Art Practice’.

⁵ RGB Colour Model: a colour (seen as light) is defined by adding together different amounts of its three primary components, Red, Green and Blue light.

⁶ Evangelina Sirgado de Sousa, screen capture of *Encoded Thoughts*, RGB version, viewed 9 September 2015, <https://vimeo.com/138700345>.

⁷ HSB Colour Model: a colour is a combination of a Hue with some degree of Saturation and Brightness in different percentages.

⁸ Evangelina Sirgado de Sousa, screen capture of *Encoded Thoughts*, HSB version, viewed 9 September 2015, <https://vimeo.com/138637189>.

⁹ MIDI (Musical Instrument Digital Interface) is a protocol that allows digital musical tools to communicate with each other. Among other advantages, it creates a correlation between standard musical notation and the so-called MIDI notes. MIDI note 60, for instance, corresponds to mid C (C4).

¹⁰ It was the author’s intention to insert in here the actual Processing code for this mapping. This wasn’t possible but the author can be contacted for further discussion of the algorithm.

¹¹ Isaac Newton, 'The Project Gutenberg eBook of Opticks', eBook #33504, 155, viewed 2 November 2015,

<http://www.gutenberg.org/files/33504/33504-h/33504-h.htm>.

¹² Jeffrey N. Howard, 'Harmonic Content Influence on Colour-Choice Association with Unaccompanied Tones', *JMM: The Journal of Music and Meaning* 10 (2012): 45-78.

¹³ Isaac Newton, 'The Project Gutenberg eBook of Opticks'.

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